

Table 4-7b. Sample Health Incidence Dataset

Endpoint Group	Endpoint	Year	Race	Gender	Ethnicity	Start Age	End Age	Column	Row	Type	Value
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	2	Incidence	2.62E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	3	Incidence	7.73E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	4	Incidence	1.68E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	5	Incidence	5.45E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	6	Incidence	1.02E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	7	Incidence	2.72E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	8	Incidence	5.05E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	9	Incidence	3.03E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	10	Incidence	5.89E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	11	Incidence	3.11E-06
Hospital Admissions, Respiratory	HA, Asthma	2002				0	99	1	12	Incidence	3.30E-06

4.1.5 Population Data

The population data is used to estimate population exposure and in turn any adverse health effects associated with a change in air pollution. BenMAP-CE allows you to specify population subgroups, as well as the year of the population estimate.

Population data loaded into BenMAP-CE must be associated with a **Population Configuration**, which defines the population subgroups present in the data. Race, ethnicity, and gender are unique text values representing population subgroups. Age ranges are defined by integer values for starting age and ending age (inclusive), and a unique text value representing the name of the age range. For example, '0T01' might be used as a name for the age range defined by a start age of zero and an end age of one, thus consisting of infants through the first twelve months of life and all one-year old infants. The population data provided to BenMAP-CE should then contain population values for all combinations of population subgroups specified in the associated **Population Configuration**. The population values may be non-integer values.

Fundamental Concept – Population Configuration

BenMAP-CE requires **population data** in order to estimate the adverse health effects associated with a change in air pollution. Population data may be stratified by population subgroups. To add population data in BenMAP-CE, you must first specify a **population configuration**. The population configuration is a template that specifies the categories into which your population data are organized – specifically, the population subgroups present in the population data you are planning to input. Stratifying your data by these categories is not required, but detailed population data allows you to more accurately estimate health impacts by better aligning your data with the study populations of health impact research. It also allows you to estimate and report benefits by population subgroups that may be useful to support regulatory analyses.

In the U.S. setup, BenMAP-CE uses 2020 U.S. Census block data to forecast the population in future years at various spatial resolutions. Sources of population data will vary by country and are generally available from the country's governmental agency responsible for collecting demographic and population data.

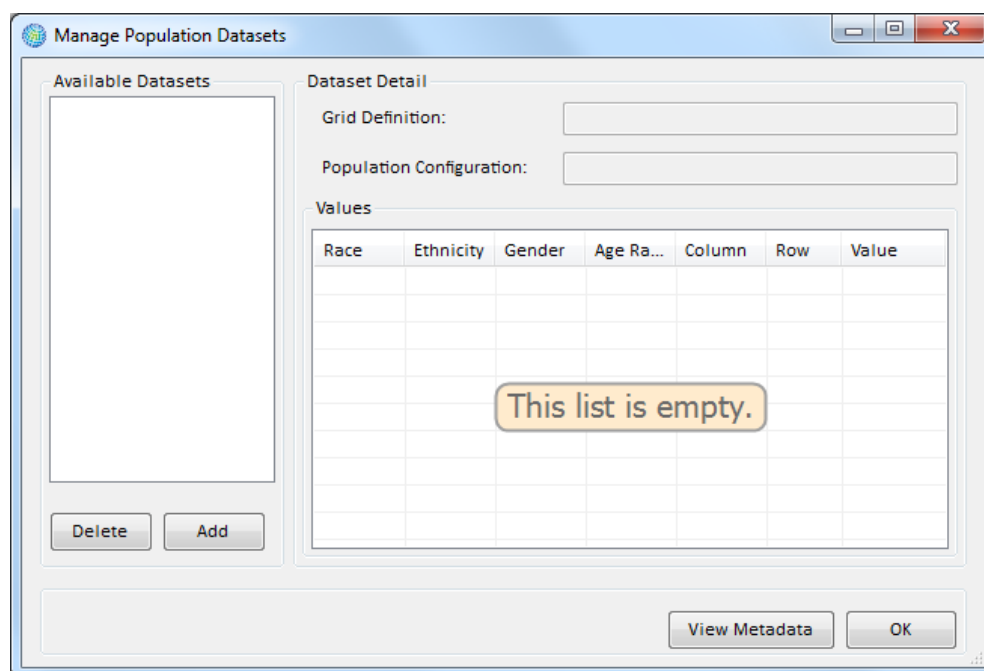
Population data must also be associated with a **Grid Definition** which specifies the geographic areas for which the data is available (see for more details the section on Grid Definitions). If population data is available for multiple grid definitions (cities and

neighborhoods, for example), you can have the option of using different sets of population data for different analyses.

BenMAP-CE can also estimate populations for Grid Definitions for which no population data is available by calculating spatial overlap percentages with Grid Definitions for which data is available.

4.1.5.1 Add Population Data

To add population data to BenMAP-CE, click on the **Manage** button below the **Population Datasets** box in the **Modify Datasets** window. The **Manage Population Datasets** window will appear.



Click on the **Add** button to display the **Load Population Dataset** window. Name the dataset using the **Population Dataset Name** box.

The **Grid Definition** drop-down list provides the list of existing grid definitions. Choose a grid definition that matches your population dataset.

The **Population Configuration** section allows you to define the variables that are in the population data file to be loaded into BenMAP-CE. Use the drop-down list to choose an existing population configuration and then view it by clicking the **View** button, or you may click the **Add** button and define a new population configuration. Clicking the **Add** button will open a **Population Configuration Definition** window where you can enter the fields that appear in the file that will be later uploaded (discussed in more detail below).

The **Browse** button to the right of the Database box allows you to find the data file that you want to load into BenMAP-CE. Once you click **Open** and load the file, the **Validate** and **View Metadata** buttons become active. You can click the **Validate** button before the file is loaded into the **Manage Population Datasets** form. BenMAP-CE's validation tool will review the file format (column names, required columns, and data types) and provide a report with any errors or warnings. You can also add metadata, which includes references and descriptions, by clicking the **View Metadata** button.

If you wish to run an analysis based on an air quality grid not already defined in BenMAP-CE, you may need to import a new population file matched to that grid definition. The PopGrid program allocates the 2020 block-level U.S. Census population to a user-defined grid, creating a population file ready for importation to BenMAP-CE.¹⁷

The *Use Population Growth Weights* checkbox should be checked when using population data generated by the PopGrid software application. The population weights file assists in forecasting population levels. See Appendix J for a more detailed discussion of population growth weights in the *United States* setup.

¹⁷ The PopGrid program may be downloaded from EPA's website. See: <http://www.epa.gov/benmap/benmap-community-edition>.

Defining a Population Configuration

If you are performing an analysis outside of the U.S. and are loading your own population data, you will need to first create a new **Population Configuration**. The Population Configuration defines the population subgroups in your population database.

Check that your population variables align with the population configuration already defined for your setup (the spellings must match exactly). If your data fails to load correctly, you will need to go back and either develop a new population configuration to match your data, or you need to revise your population database so that it matches the population configuration.

Warning

It is critical that the population subgroups defined in the population configuration match your population input data **exactly**.

To add a population configuration, click the **Add** button in the **Load Population Dataset** window.

In the **Population Configuration Name** box, replace 'PopulationConfiguration0' with a name of your choosing. Under the **Races** box click on **New** and type in the name for any races present in your population data. The names appear in both the **Races** list box and